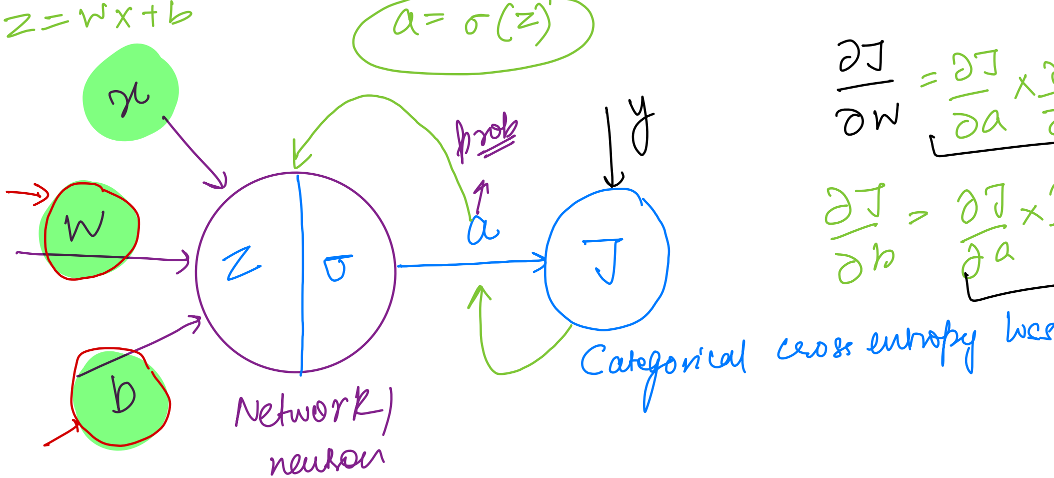


$$\frac{\partial J}{\partial w} = \underbrace{\frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z}} \times \frac{\partial z}{\partial w}$$

$$\frac{\partial J}{\partial b} = \underbrace{\frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z}} \times \frac{\partial z}{\partial b}$$

$$J = -\frac{1}{N} \sum_{i=1}^{N} y \log(a) + (1-y)\log(1-a)$$

cost funct^n

$$\text{loss} \rightarrow J = -\left(y \log(a) + (1-y)\log(1-a)\right)$$

Minimize this loss → update w & b

Backpropogation →

softmax for 2 classes

$$\frac{e^{za}}{e^{za}+e^{zb}} = \frac{\cancel{e^{za}}}{\cancel{e^{za}}}\left(\frac{1}{1+\frac{e^{zb}}{e^{za}}}\right)$$

$$\Rightarrow \frac{1}{1+\frac{e^{zb}}{e^{za}}} \Rightarrow \frac{1}{1+e^{zb-za}}$$

$$= 1/1+e^{-(za-zb)}$$

↳ z

$$w' = w - \alpha \frac{\partial J}{\partial w}$$

$\frac{\partial J}{\partial w}$ → Reduction in loss wrt w

change in J wrt change in b

$$b' = b - \alpha \frac{\partial J}{\partial b}$$

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial w}$$

$$\frac{\partial J}{\partial a} = \frac{\partial \left(-\left(y \log a + (1-y) \log(1-a)\right)\right)}{\partial a}$$

$$= -\left(y \frac{\partial \log a}{\partial a} + (1-y) \frac{\partial (\log(1-a))}{\partial a}\right)$$

$$\boxed{\frac{\partial J}{\partial a} = -\left(y\left(\frac{1}{a}\right) + (1-y) \times \frac{1}{(1-a)}(-1)\right)}$$

→ 1

$$\frac{\partial a}{\partial z} = \frac{\partial(\sigma(z))}{\partial z}$$

$\frac{\partial a}{\partial z}$ → $(1-a)\,a$ → 2

$$= \frac{\partial(\sigma(z))}{\partial z} = \frac{\partial\left(1/1+e^{-z}\right)}{\partial z}$$

$$= \frac{\partial}{\partial z}\left(1+e^{-z}\right)^{-1} = -\left(1+e^{-z}\right)^{-2} \times -e^{-z}$$

$$= \frac{1}{\left(1+e^{-z}\right)^2}\left(-e^{-z}\right) \Rightarrow \frac{1}{\left(1+e^{-z}\right)} \times \frac{\left(e^{-z}\right)}{\left(1+e^{-z}\right)}$$

$$\Rightarrow \frac{1}{\left(1+e^{-z}\right)} \times \frac{e^{-z}+1+1}{\left(1+e^{-z}\right)} \Rightarrow \left(\frac{1}{1+e^{-z}}\right) \times \left(1 - \frac{1}{1+e^{-z}}\right) = \left(1-\sigma(z)\right)\sigma(z)$$

sig, sig

$$\Rightarrow (1-a)\,a$$

$$\frac{\partial z}{\partial w} \quad 1 \quad x w + b \quad 0$$

$$= x \rightarrow 3$$

$$\frac{\partial z}{\partial b} = 1$$

$$\frac{\partial J}{\partial w} = ① \times ② \times ③ \Rightarrow \left[\frac{-y}{a} + \frac{(1-y)}{\cancel{(1-a)}}\right] \times a(1-a) \times x$$

$$\frac{-y(1-a) + (1-y)a}{\cancel{a(1-a)}} \times \cancel{a(1-a)} \times x$$

$$\frac{\partial z}{\partial w}$$

$$\Rightarrow -y + \cancel{ya} + a - \cancel{ya} \Rightarrow (a-y)\,x$$

$$\boxed{\frac{\partial J}{\partial w} = (a-y) \times x}$$

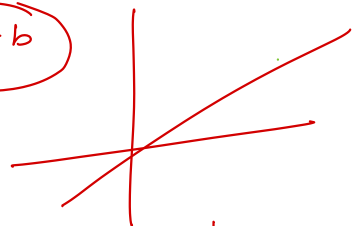
$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial w}$$

$$\boxed{\frac{\partial J}{\partial b} = (a-y) \times 1}$$

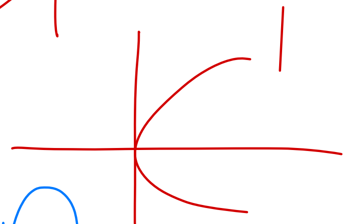
$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial a} \times \frac{\partial a}{\partial z} \times \frac{\partial z}{\partial b}$$

Non-linear decision boundary

1. $ax+b$

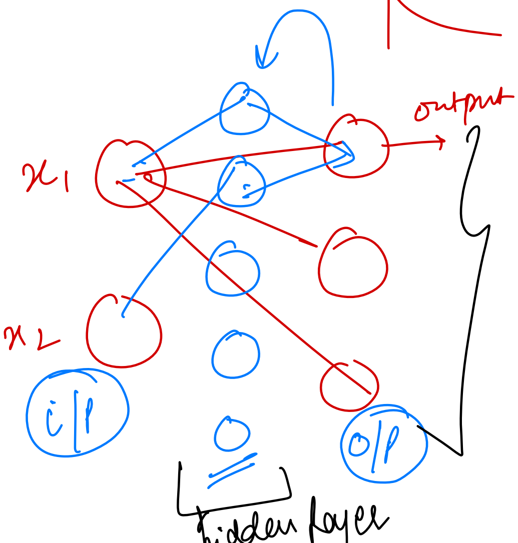


2. ax^2+bx+c

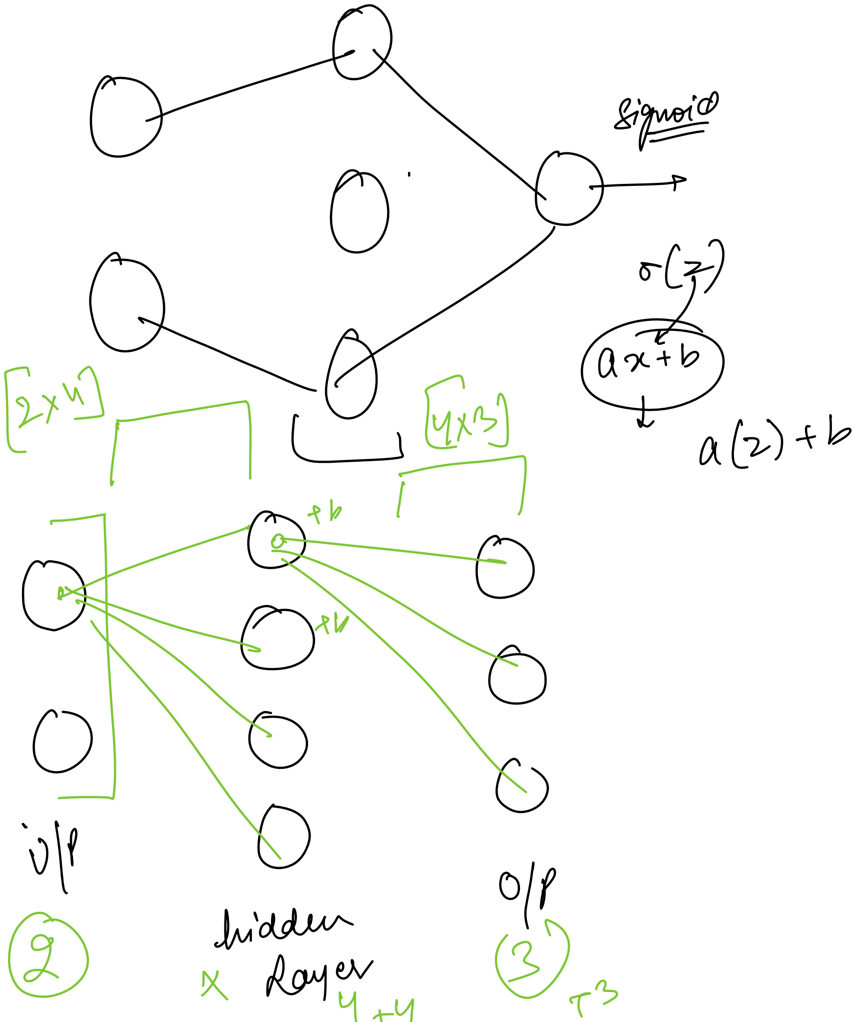


1st way

3.



Adding a hidden [layer]
↳ can induce non-linearity

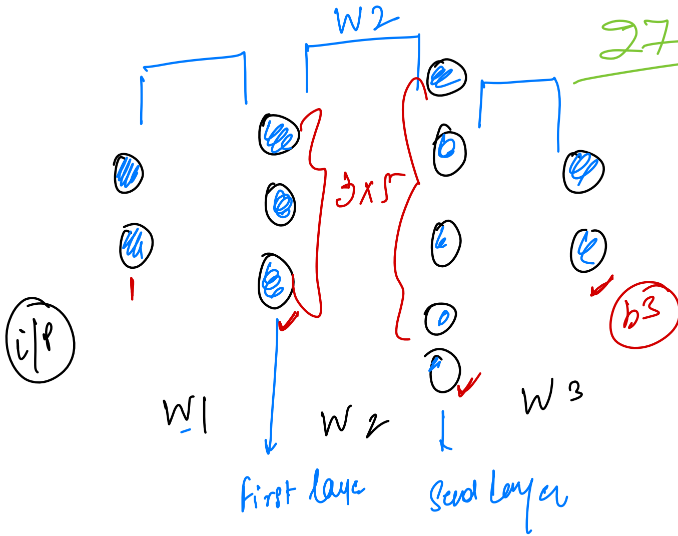


Sigmoid
σ(z)
ax+b
a(z)+b
[2x4]
[4x3]
+b
+b
i/p
hidden
Layer
o/p
2
3
4
+4
+3

Weights 8 + 12 → 20

4 3 → 7

27 parameters.



1x4

[1x2]